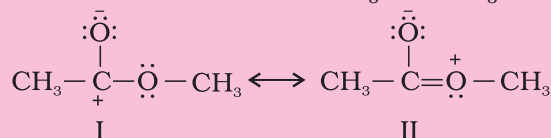


Problem 12.18

Explain why the following two structures, I and II cannot be the major contributors to the real structure of $\text{CH}_3\text{COOCH}_3$.

**Solution**

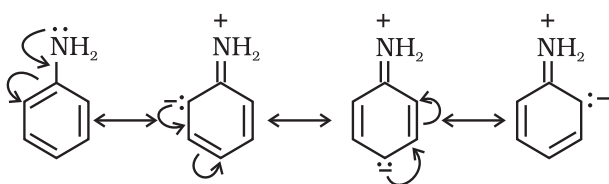
The two structures are less important contributors as they involve charge separation. Additionally, structure I contains a carbon atom with an incomplete octet.

12.7.7 Resonance Effect

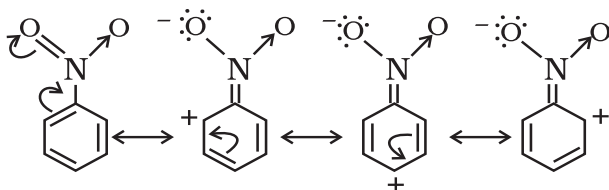
The resonance effect is defined as 'the polarity produced in the molecule by the interaction of two π -bonds or between a π -bond and lone pair of electrons present on an adjacent atom'. The effect is transmitted through the chain. There are two types of resonance or mesomeric effect designated as R or M effect.

(i) Positive Resonance Effect (+R effect)

In this effect, the transfer of electrons is away from an atom or substituent group attached to the conjugated system. This electron displacement makes certain positions in the molecule of high electron densities. This effect in aniline is shown as :

**(ii) Negative Resonance Effect (-R effect)**

This effect is observed when the transfer of electrons is towards the atom or substituent group attached to the conjugated system. For example in nitrobenzene this electron displacement can be depicted as :



The atoms or substituent groups, which represent +R or -R electron displacement effects are as follows :

+R effect: - halogen, -OH, -OR, -OCOR, -NH₂, -NHR, -NR₂, -NHCOR,

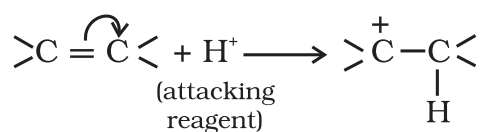
- R effect: - COOH, -CHO, >C=O, - CN, -NO₂

The presence of alternate single and double bonds in an open chain or cyclic system is termed as a conjugated system. These systems often show abnormal behaviour. The examples are 1,3-butadiene, aniline and nitrobenzene etc. In such systems, the π -electrons are delocalised and the system develops polarity.

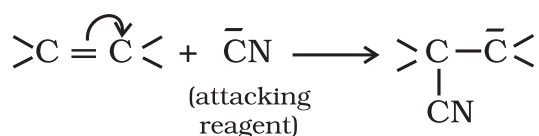
12.7.8 Electromeric Effect (E effect)

It is a temporary effect. The organic compounds having a **multiple bond** (a double or triple bond) show this effect in the presence of an attacking reagent only. It is defined as the complete transfer of a shared pair of π -electrons to one of the atoms joined by a multiple bond on the demand of an attacking reagent. The effect is annulled as soon as the attacking reagent is removed from the domain of the reaction. It is represented by E and the shifting of the electrons is shown by a curved arrow ($\curvearrowright$). There are two distinct types of electromeric effect.

(i) Positive Electromeric Effect (+E effect) In this effect the π -electrons of the multiple bond are transferred to that atom to which the reagent gets attached. For example :



(ii) Negative Electromeric Effect (-E effect) In this effect the π -electrons of the multiple bond are transferred to that atom to which the attacking reagent does not get attached. For example:



When inductive and electromeric effects operate in opposite directions, the electromeric effect predominates.

12.7.9 Hyperconjugation

Hyperconjugation is a general stabilising interaction. It involves delocalisation of σ electrons of C—H bond of an alkyl group directly attached to an atom of unsaturated system or to an atom with an unshared p orbital. The σ electrons of C—H bond of the alkyl group enter into partial conjugation with the attached unsaturated system or with the unshared p orbital. Hyperconjugation is a permanent effect.

To understand hyperconjugation effect, let us take an example of CH_3CH_2^+ (ethyl cation) in which the positively charged carbon atom has an empty p orbital. One of the C—H bonds of the methyl group can align in the plane of this empty p orbital and the electrons constituting the C—H bond in plane with this p orbital can then be delocalised into the empty p orbital as depicted in Fig. 12.4 (a).

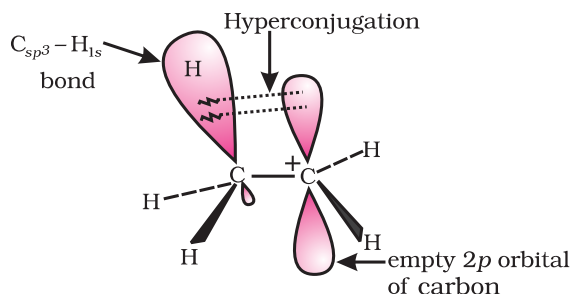
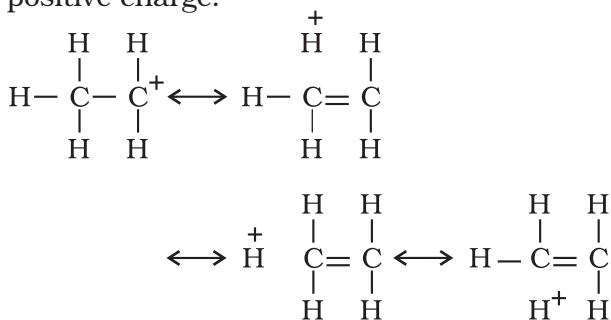
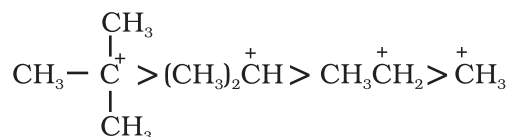


Fig. 12.4(a) Orbital diagram showing hyperconjugation in ethyl cation

This type of overlap stabilises the carbocation because electron density from the adjacent σ bond helps in dispersing the positive charge.



In general, greater the number of alkyl groups attached to a positively charged carbon atom, the greater is the hyperconjugation interaction and stabilisation of the cation. Thus, we have the following relative stability of carbocations :



Hyperconjugation is also possible in alkenes and alkylarenes.

Delocalisation of electrons by hyperconjugation in the case of alkene can be depicted as in Fig. 12.4(b).

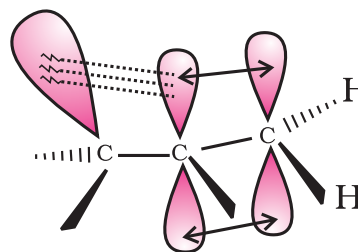
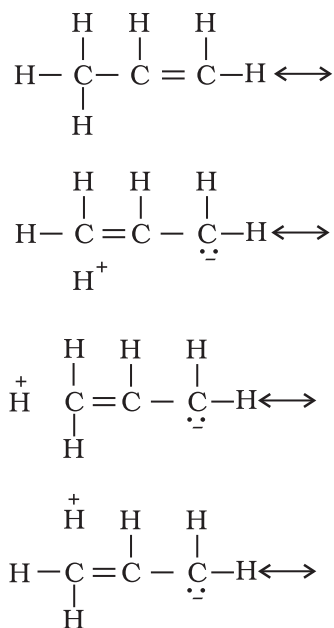


Fig. 12.4(b) Orbital diagram showing hyperconjugation in propene

There are various ways of looking at the hyperconjugative effect. One of the way is to regard C—H bond as possessing partial ionic character due to resonance.



The hyperconjugation may also be regarded as no bond resonance.

Problem 12.19

Explain why $(\text{CH}_3)_3\text{C}^+$ is more stable than CH_3CH_2^+ and CH_3^+ is the least stable cation.

Solution

Hyperconjugation interaction in $(\text{CH}_3)_3\text{C}^+$ is greater than in CH_3CH_2^+ as the $(\text{CH}_3)_3\text{C}^+$ has nine C-H bonds. In CH_3^+ , vacant p orbital is perpendicular to the plane in which C-H bonds lie; hence cannot overlap with it. Thus, CH_3^+ lacks hyperconjugative stability.

12.7.10 Types of Organic Reactions and Mechanisms

Organic reactions can be classified into the following categories:

- (i) Substitution reactions
- (ii) Addition reactions
- (iii) Elimination reactions
- (iv) Rearrangement reactions

You will be studying these reactions in Unit 13 and later in class XII.

12.8 METHODS OF PURIFICATION OF ORGANIC COMPOUNDS

Once an organic compound is extracted from a natural source or synthesised in the laboratory, it is essential to purify it. Various methods used for the purification of organic compounds are based on the nature of the compound and the impurity present in it.

The common techniques used for purification are as follows :

- (i) Sublimation
- (ii) Crystallisation
- (iii) Distillation
- (iv) Differential extraction and
- (v) Chromatography

Finally, the purity of a compound is ascertained by determining its melting or boiling point. Most of the pure compounds have sharp melting points and boiling points.

New methods of checking the purity of an organic compound are based on different types of chromatographic and spectroscopic techniques.

12.8.1 Sublimation

You have learnt earlier that on heating, some solid substances change from solid to vapour state without passing through liquid state. The purification technique based on the above principle is known as sublimation and is used to separate sublimable compounds from non-sublimable impurities.

12.8.2 Crystallisation

This is one of the most commonly used techniques for the purification of solid organic compounds. It is based on the difference in the solubilities of the compound and the impurities in a suitable solvent. The impure compound is dissolved in a solvent in which it is sparingly soluble at room temperature but appreciably soluble at higher temperature. The solution is concentrated to get a nearly saturated solution. On cooling the solution, pure compound crystallises out and is removed by filtration. The filtrate (mother liquor) contains impurities and small quantity of the compound. If the compound is highly soluble in one solvent and very little soluble in another solvent, crystallisation can be satisfactorily carried out in a mixture of these solvents. Impurities, which impart colour to the solution are removed by adsorbing over activated charcoal. Repeated crystallisation becomes necessary for the purification of compounds containing impurities of comparable solubilities.

12.8.3 Distillation

This important method is used to separate (i) volatile liquids from nonvolatile impurities and (ii) the liquids having sufficient difference in their boiling points. Liquids having different boiling points vaporise at different temperatures. The vapours are cooled and the liquids so formed are collected separately. Chloroform (b.p. 334 K) and aniline (b.p. 457 K) are easily separated by the technique of distillation (Fig 12.5). The liquid mixture is

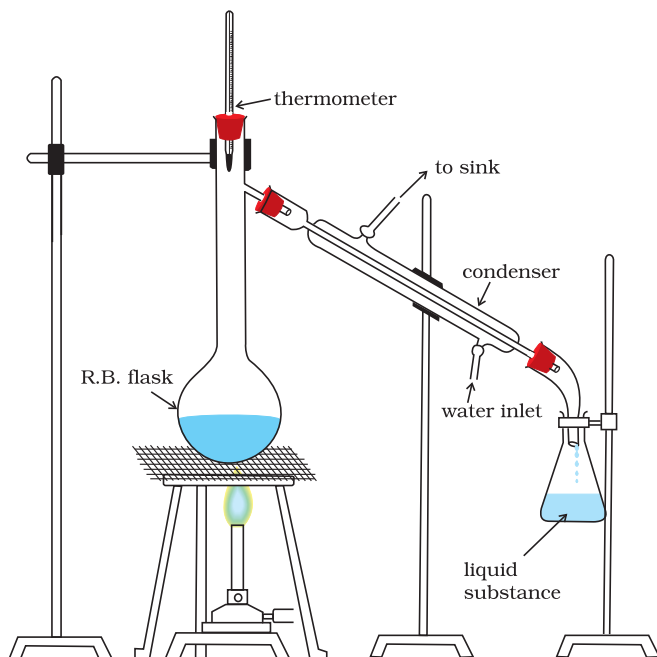


Fig.12.5 Simple distillation. The vapours of a substance formed are condensed and the liquid is collected in conical flask.

taken in a round bottom flask and heated carefully. On boiling, the vapours of lower boiling component are formed first. The vapours are condensed by using a condenser and the liquid is collected in a receiver. The vapours of higher boiling component form later and the liquid can be collected separately.

Fractional Distillation: If the difference in boiling points of two liquids is not much, simple distillation cannot be used to separate them. The vapours of such liquids are formed within the same temperature range and are condensed simultaneously. The technique of fractional distillation is used in such cases. In this technique, vapours of a liquid mixture are passed through a fractionating column before condensation. The fractionating column is fitted over the mouth of the round bottom flask (Fig.12.6).

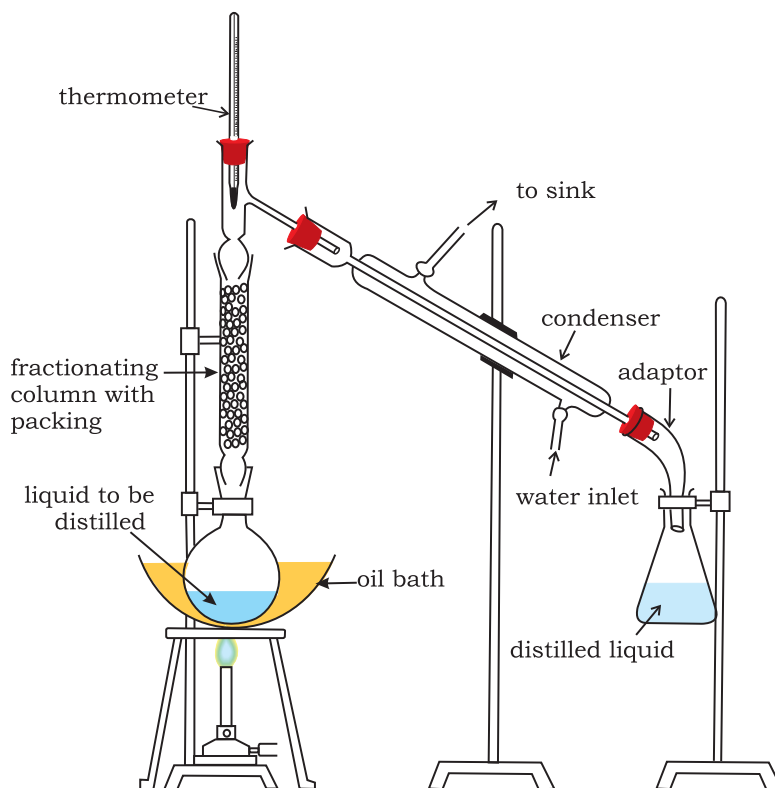


Fig.12.6 Fractional distillation. The vapours of lower boiling fraction reach the top of the column first followed by vapours of higher boiling fractions.